

Vectors Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 E, F, G and H are the midpoints of AD, DC, CB and BA respectively. Let $\overrightarrow{BA} = \underline{u}$, $\overrightarrow{AD} = \underline{v}$, $\overrightarrow{DC} = \underline{w}$

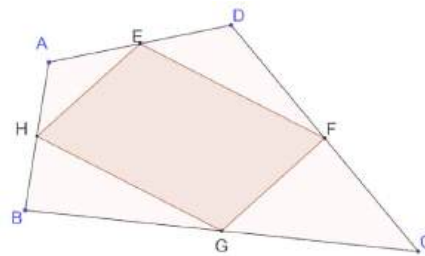
and $\overrightarrow{CB} = \underline{z}$. $\overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DC} + \overrightarrow{CB} = \overrightarrow{BB} = \underline{0}$

(i). Explain why $\underline{u} + \underline{v} + \underline{w} + \underline{z} = \underline{0}$.

(ii). Show that $\overrightarrow{HE} = \frac{1}{2}(\underline{u} + \underline{v})$.

$$\overrightarrow{HA} + \overrightarrow{AE} = \overrightarrow{HE} = \frac{1}{2}\underline{u} + \frac{1}{2}\underline{v} = \frac{1}{2}(\underline{u} + \underline{v})$$

(iii). Show that $\overrightarrow{GF} = -\frac{1}{2}(\underline{w} + \underline{z})$. $\overrightarrow{GF} = \overrightarrow{GC} + \overrightarrow{CF} = -\frac{1}{2}\underline{z} + -\frac{1}{2}\underline{w} = -\frac{1}{2}(\underline{w} + \underline{z})$



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- 2 Find the angle between the vectors $\underline{a} = \begin{pmatrix} 4 \\ -5 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$, correct to the nearest degree.

Angle between $\underline{a} = \begin{pmatrix} 4 \\ -5 \end{pmatrix}$ $\underline{b} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$ (nearest degree)

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta$$

$$|\underline{a}| = \sqrt{4^2 + (-5)^2} = \sqrt{41} \quad |\underline{b}| = \sqrt{(-2)^2 + 4^2} = \sqrt{20}$$

$$\underline{a} \cdot \underline{b} = \sqrt{41} \times \sqrt{20} \cos \theta$$

Also $\underline{a} \cdot \underline{b} = -8 - 20 = -28$

$$\cos \theta = \frac{-28}{\sqrt{820}} \quad \theta = 167.54^\circ$$

- 3 Six equilateral triangles form a hexagon with side lengths of 4 cm. The

vectors $\underline{u}$ and $\underline{v}$ and $\underline{w}$ are shown in the diagram. What is the value

of $\underline{u} \cdot (\underline{u} + \underline{v} + \underline{w})$?

$$\underline{u} + \underline{v} + \underline{w} = 2\underline{v}$$

$$\therefore \underline{u} \cdot (\underline{u} + \underline{v} + \underline{w}) = \underline{u} \cdot 2\underline{v}$$

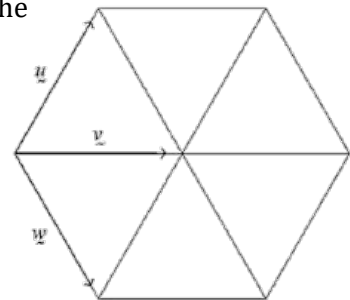
$$\cos \theta = \frac{\underline{u} \cdot 2\underline{v}}{|\underline{u}| |2\underline{v}|}$$

$$\therefore \underline{u} \cdot 2\underline{v} = |\underline{u}| |2\underline{v}| \cos \theta$$

Since all the triangles are equilateral,

$$\theta = \frac{\pi}{3}, |\underline{u}| = 4 \text{ and } |2\underline{v}| = 8.$$

$$\therefore \underline{u} \cdot 2\underline{v} = 4 \times 8 \times \cos \frac{\pi}{3} = 16$$



2

- 4 Resolve the vector $\underline{a}$ into component form $\underline{a} = x\underline{i} + y\underline{j}$, given $\underline{a}$ has a magnitude of 6 units and

has a direction of 50° to the positive x - axis. Give answers correct to two decimal places.

$$|\underline{a}| = 6 \text{ and } \theta = 50^\circ$$

$$\underline{a} = |\underline{a}| \cos \theta \underline{i} + |\underline{a}| \sin \theta \underline{j}$$

$$= 6 \cos(50^\circ) \underline{i} + 6 \sin(50^\circ) \underline{j}$$

$$= 3.86 \underline{i} + 4.60 \underline{j}$$

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- 5 Given $\underline{c} = 3\underline{i} - 6\underline{j}$, find vector $\underline{d}$ of magnitude 5 in the direction of $\underline{c}$.

2

Find the magnitude of the vector:

$$|\underline{c}| = \sqrt{3^2 + (-6)^2} = \sqrt{45} = 3\sqrt{5}$$

Find the unit vector:

$$\hat{\underline{c}} = \frac{1}{3\sqrt{5}}(3\underline{i} - 6\underline{j}) = \frac{\sqrt{5}}{5}(\underline{i} - 2\underline{j})$$

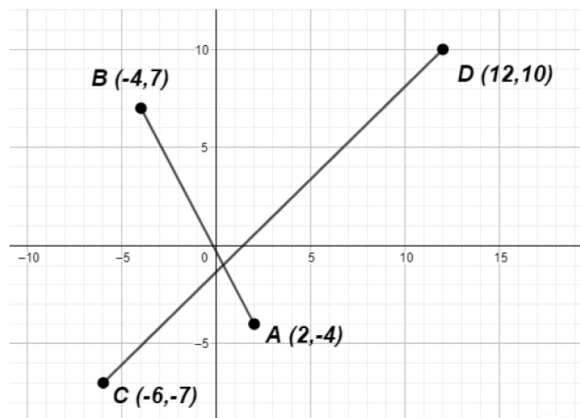
Multiply the unit vector in the direction required.

$$\underline{d} = \frac{5\sqrt{5}}{5}(\underline{i} - 2\underline{j}) = \sqrt{5}(\underline{i} - 2\underline{j})$$

- 6 (i). Explain in words only why $\underline{i} \cdot \underline{j} = 0$?

The unit vector $\underline{i}$ lies along the x -axis, whilst the unit vector $\underline{j}$ lies along the y -axis. As such, the two unit vectors are perpendicular, resulting in the dot product of $\underline{i} \cdot \underline{j} = 0$

- (ii). The diagram below shows the lines AB and CD. The line AB passes through the points $(2, -4)$ and $(-4, 7)$ and the line CD passes through $(-6, -7)$ and $(12, 10)$.



Use vector methods to find the acute angle between the two lines. Write your answer to the nearest degree.

$$\vec{AB} = (4, 7) - (2, -4) = (-6, 11)$$

$$\vec{CD} = (12, 10) - (-6, -7) = (18, 17)$$

$$\cos\theta = \frac{\vec{AB} \cdot \vec{CD}}{|\vec{AB}| |\vec{CD}|}$$

$$\cos\theta = \frac{-6(18) + 11(17)}{(\sqrt{(-6)^2 + 11^2})(\sqrt{18^2 + 17^2})}$$

$$\theta = 75^\circ \text{ (nearest degree)}$$

- 7 Which of the following vectors is perpendicular to $\begin{pmatrix} 4 \\ -1 \end{pmatrix}$ and has a magnitude of 5?

2

$$\text{Magnitude of given vector} = \sqrt{4^2 + (-1)^2} = \sqrt{17}$$

For a vector perpendicular to $\begin{pmatrix} 4 \\ -1 \end{pmatrix}$, we need a vector $\begin{pmatrix} a \\ b \end{pmatrix}$ such that the dot product,

$$\begin{pmatrix} 4 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = 0 \quad \therefore \text{unit vector perpendicular to given vector} = \frac{1}{\sqrt{17}} \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

$$\therefore \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \quad \text{To have a magnitude 5, required solution is } 5 \times \frac{1}{\sqrt{17}} \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

- 8 Suppose $\underline{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, $\underline{v} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$ and θ is the acute angle between them. Show that the exact value of $\sin 2\theta$ is $\frac{4}{5}$. Give clear reasoning for your answer.

4

Consider:

$$\cos\theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} = \frac{2 \times 0 + 1 \times 3}{\sqrt{2^2 + 1^2} \times \sqrt{0^2 + 3^2}} = \frac{3}{\sqrt{5} \times \sqrt{9}} = \frac{1}{\sqrt{5}}$$

Using the identity $\sin^2\theta + \cos^2\theta = 1$, we have:

$$\sin^2\theta + \left(\frac{1}{\sqrt{5}}\right)^2 = 1 \quad \sin^2\theta = \frac{4}{5}$$

$$\sin\theta = \frac{2}{\sqrt{5}}, \text{ where } \sin\theta > 0 \text{ since } \theta \text{ is acute.}$$

$$\text{Using } \sin 2\theta = 2 \sin\theta \cos\theta, = 2 \times \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{5}} = \frac{4}{5}$$

9 Consider the vectors $\overrightarrow{OP} = -2\mathbf{i} + \mathbf{j}$ and $\overrightarrow{OQ} = 4\mathbf{i} - 3\mathbf{j}$.

(i). Find the vector $\mathbf{u} = \overrightarrow{PQ}$.

$$\begin{aligned}\mathbf{u} &= \overrightarrow{PQ} \\ &= \overrightarrow{PO} + \overrightarrow{OQ} \\ &= -(\overrightarrow{OP}) + \overrightarrow{OQ} \\ &= -(-2\mathbf{i} + \mathbf{j}) + 4\mathbf{i} - 3\mathbf{j} \\ &= 2\mathbf{i} - \mathbf{j} + 4\mathbf{i} - 3\mathbf{j} \\ &= 6\mathbf{i} - 4\mathbf{j} \quad [1]\end{aligned}$$

(ii). Given that $\mathbf{v} = -\mathbf{i} + 2\mathbf{j}$, find $\mathbf{u} \cdot \mathbf{v}$.

$$\begin{aligned}\mathbf{u} \cdot \mathbf{v} &= (6\mathbf{i} - 4\mathbf{j}) \cdot (-\mathbf{i} + 2\mathbf{j}) \\ &= 6 \times (-1) + (-4) \times 2 \\ &= -14 \quad [1]\end{aligned}$$

(iii). Find $\text{proj}_{\mathbf{u}} \mathbf{v}$.

$$\begin{aligned}\text{proj}_{\mathbf{u}} \mathbf{v} &= \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}|^2} \mathbf{u} \\ &= \frac{-14}{(6^2 + 4^2)} (6\mathbf{i} - 4\mathbf{j}) \\ &= \frac{-14}{52} (6\mathbf{i} - 4\mathbf{j}) \\ &= \frac{-84}{52} \mathbf{i} + \frac{56}{52} \mathbf{j} \\ &= -\frac{21}{13} \mathbf{i} + \frac{14}{13} \mathbf{j} \quad [2]\end{aligned}$$

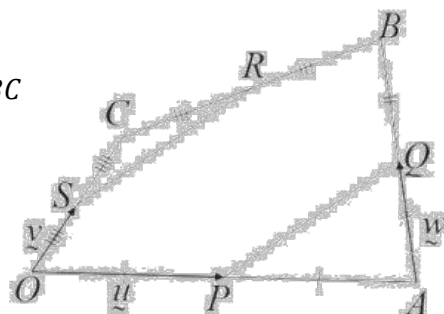
10 Consider the diagram below where $OABC$ is a quadrilateral and P, Q, R and S are the midpoints of the intervals OA, AB, BC and CO , respectively. Let $\overrightarrow{OP} = \mathbf{u}$, $\overrightarrow{OS} = \mathbf{v}$ and $\overrightarrow{AQ} = \mathbf{w}$.

Prove that $\overrightarrow{PQ} = \overrightarrow{SR}$.

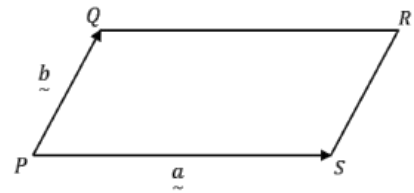
$$\begin{aligned}\overrightarrow{PQ} &= \overrightarrow{PA} + \overrightarrow{AQ} \\ &= \frac{1}{2} \overrightarrow{OA} + \overrightarrow{AQ} \\ &= \overrightarrow{OP} + \overrightarrow{AQ} \\ &= \mathbf{u} + \mathbf{w}\end{aligned}$$

$$\begin{aligned}\overrightarrow{SR} &= \overrightarrow{SC} + \overrightarrow{CR} \\ &= \frac{1}{2} \overrightarrow{OC} + \frac{1}{2} \overrightarrow{CB} \\ &= \overrightarrow{OS} + \frac{1}{2} (\overrightarrow{CO} + \overrightarrow{OA} + \overrightarrow{AB}) \\ &= \mathbf{v} + \frac{1}{2} (-2\mathbf{v} + 2\mathbf{u} + 2\mathbf{w}) \\ &= \mathbf{v} - \mathbf{v} + \mathbf{u} + \mathbf{w} \\ &= \mathbf{u} + \mathbf{w}\end{aligned}$$

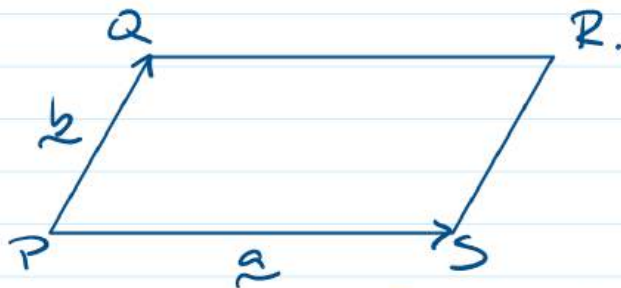
$$\therefore \overrightarrow{PQ} = \overrightarrow{SR}$$



- 11 $PQRS$ is a parallelogram, where $\overrightarrow{PQ} = \underline{u}$ and $\overrightarrow{PS} = \underline{b}$. Prove, using vectors, that the sum of the squares of the lengths of the diagonals is equal to the sum of the squares of the lengths of the sides.



2



$$\begin{aligned}\overrightarrow{QR} &= \overrightarrow{PS} = \underline{a} \\ \overrightarrow{SR} &= \overrightarrow{PQ} = \underline{b}\end{aligned}$$

diagonals:
$$\begin{aligned}\overrightarrow{PR} &= \underline{a} + \underline{b} \\ \overrightarrow{QS} &= \underline{a} - \underline{b}\end{aligned}$$

$$\begin{aligned}|\underline{a} + \underline{b}|^2 + |\underline{a} - \underline{b}|^2 &= (\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b}) + (\underline{a} - \underline{b}) \cdot (\underline{a} - \underline{b}) \\ &= \underline{a} \cdot \underline{a} + \cancel{2\underline{a} \cdot \underline{b}} + \underline{b} \cdot \underline{b} + \underline{a} \cdot \underline{a} - \cancel{2\underline{a} \cdot \underline{b}} + \underline{b} \cdot \underline{b} \\ &= |\underline{a}|^2 + |\underline{b}|^2 + |\underline{a}|^2 + |\underline{b}|^2\end{aligned}$$

which is the sum of the squares of the sides

Vectors Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 For what values of m are the two distinct vectors $\begin{pmatrix} m \\ 2m+6 \end{pmatrix}$ and $\begin{pmatrix} m+1 \\ -1 \end{pmatrix}$ perpendicular? 2

perpendicular when

$$\begin{pmatrix} m \\ 2m+6 \end{pmatrix} \cdot \begin{pmatrix} m+1 \\ -1 \end{pmatrix} = 0$$

$$m(m+1) + -1(2m+6) = 0$$

$$m^2 + m - 2m - 6 = 0$$

$$m^2 - m - 6 = 0$$

$$(m-3)(m+2) = 0$$

$$\therefore m = 3 \text{ or } -2$$

- 2 The vectors $u = \begin{pmatrix} a \\ 3 \end{pmatrix}$ and $v = \begin{pmatrix} 2 \\ 7 \end{pmatrix}$ are parallel. Find the value of a . 1

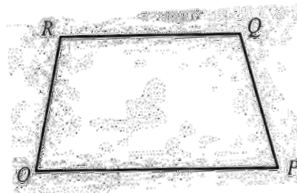
$$u = \begin{pmatrix} a \\ 3 \end{pmatrix} \quad v = \begin{pmatrix} 2 \\ 7 \end{pmatrix}$$

For parallel $\begin{pmatrix} a \\ 3 \end{pmatrix} = k \begin{pmatrix} 2 \\ 7 \end{pmatrix}$

$$3 = k \times 7 \quad \therefore a = \frac{3}{7} \times 2$$

$$\frac{3}{7} = k \quad a = \frac{6}{7}$$

- 3 The diagram shows a trapezium $OPQR$. If $\overrightarrow{OP} = p$, $\overrightarrow{OR} = r$ and using that fact that $\overrightarrow{OP} = 3\overrightarrow{RQ}$, find an expression for $\overrightarrow{PQ}$ in terms of p and r . 1



$$\begin{aligned} \overrightarrow{PQ} &= \overrightarrow{PO} + \overrightarrow{OR} + \overrightarrow{RQ} \\ &= -p + r + \frac{1}{3}p \\ &= r - \frac{2}{3}p \end{aligned}$$

- 4 Two vectors are given by $a = 3i - mj$ and $b = -2i + nj$, where $m > 0$. If $|a| = 5$ and a is perpendicular to b , find the values of m and n . 2

$$a = 3i - mj \quad b = -2i + nj$$

$$|a| = 5 \quad a \perp b \Rightarrow a \cdot b = 0$$

$$\sqrt{3^2 + m^2} = 5 \quad \therefore 3(-2) - 4(n) = 0$$

$$m^2 = 16 \quad 4n = -6$$

$$m = 4 \quad n = -\frac{3}{2}$$

- 5 What is the projection of the force $F = ai + bj$, where a and b are real constants, in the direction of $w = i + j$? 2

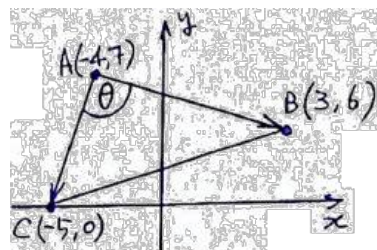
$$\hat{w} = \frac{1}{\sqrt{2}}(i + j)$$

$$\text{proj}_w F = (F \cdot \hat{w}) \hat{w}$$

$$= \left((a+b) \cdot \frac{1}{\sqrt{2}}(i + j) \right) \frac{1}{\sqrt{2}}(i + j) = \frac{1}{2}(a+b)(i + j)$$

$$= \frac{1}{2}(a+b)w$$

6 Triangle ABC has vertices $A(-4, 7)$, $B(3, 6)$ and $C(-5, 0)$.



(i). Express $\vec{AB}$ and $\vec{AC}$ in column vector form.

$$\begin{aligned} i) \quad \vec{AB} &= \vec{OB} - \vec{OA} & \vec{AC} &= \vec{OC} - \vec{OA} \\ &= \begin{pmatrix} 3 \\ 6 \end{pmatrix} - \begin{pmatrix} -4 \\ 7 \end{pmatrix} & &= \begin{pmatrix} -5 \\ 0 \end{pmatrix} - \begin{pmatrix} -4 \\ 7 \end{pmatrix} \\ &= \begin{pmatrix} 7 \\ -1 \end{pmatrix} \quad (1) & &= \begin{pmatrix} -1 \\ -7 \end{pmatrix} \quad (1) \end{aligned}$$

(ii). Find the magnitude of $\vec{AB}$ and $\vec{AC}$.

$$\begin{aligned} ii) \quad |\vec{AB}| &= \sqrt{7^2 + (-1)^2} & |\vec{AC}| &= \sqrt{(-1)^2 + (-7)^2} \\ &= \sqrt{50} & &= \sqrt{50} \\ &= 5\sqrt{2} \quad (1) & &= 5\sqrt{2} \quad (1) \end{aligned}$$

(iii). Find $\angle BAC$.

$$\begin{aligned} iii) \quad \vec{AB} \cdot \vec{AC} &= |\vec{AB}| |\vec{AC}| \cos \theta \\ -7 + 7 &= 5\sqrt{2} \times 5\sqrt{2} \times \cos \theta \quad (1) \\ 0 &= 50 \cos \theta \\ \cos \theta &= 0 \\ \theta &= \frac{\pi}{2} \quad (\theta \text{ is angle in the triangle, } \theta < \pi) \\ \therefore \angle BAC &= \frac{\pi}{2} \text{ radians} \quad (1) \end{aligned}$$

(iv). Hence calculate the area of the triangle.

$$\begin{aligned} iv) \quad \text{Area} &= \frac{1}{2}bh \quad (\text{since } \angle BAC = 90^\circ) \\ &= \frac{1}{2} \times 5\sqrt{2} \times 5\sqrt{2} \\ &= 25 \text{ units}^2 \quad (1) \end{aligned}$$

7 Vector $\vec{AB}$ has a magnitude of $2\sqrt{3}$ and makes an angle of 150° with the positive y -axis, measuring anticlockwise from the positive y -axis. Write a unit vector in the direction of $\vec{AB}$ in the form $a\hat{i} + b\hat{j}$, where a and b are constants written in simplest exact form.

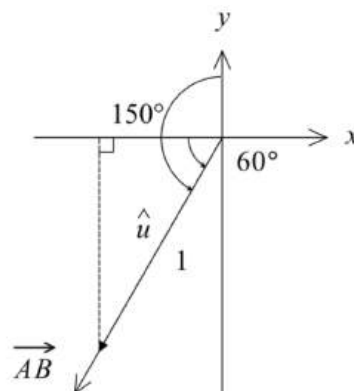
Let $\hat{u}$ be the unit vector of $\vec{AB}$.

Then, $\hat{u}$ points in the same direction as $\vec{AB}$ and has a magnitude of 1 unit.

150° anticlockwise from the positive y axis is $150^\circ - 90^\circ = 60^\circ$ in the third quadrant, where $\cos \theta < 0$ and $\sin \theta < 0$.

Hence, by right angle trig:

$$\begin{aligned} \hat{u} &= (-\cos 60^\circ)\hat{i} + (-\sin 60^\circ)\hat{j} \\ &= -\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j} \end{aligned}$$



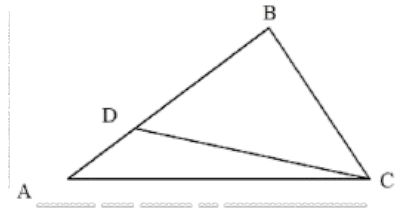
8 Find the angle between the vectors $u = -2\hat{i} + 6\hat{j}$ and $v = 4\hat{i} - 2\hat{j}$.

$$\begin{aligned} \cos \theta &= \frac{u \cdot v}{|u| |v|} = \frac{-2 \times 4 + 6 \times -2}{\sqrt{(-2)^2 + 6^2} \times \sqrt{4^2 + (-2)^2}} \\ &= \frac{-20}{2\sqrt{10} \times 2\sqrt{5}} = \frac{-20}{20\sqrt{2}} \\ \cos \theta &= -\frac{1}{\sqrt{2}} \quad \frac{\sqrt{5}A}{\sqrt{1}C} \\ \theta &= 180 - 45 \text{ or } 180 + 45 \\ \theta &= 135^\circ \text{ or } 225^\circ \\ \theta &= 135^\circ \text{ as angle between lines } < 180^\circ \\ &\quad (180 - 135 = 45^\circ \text{ is also correct}) \end{aligned}$$

- 9 In $\triangle ABC$, D is a point on AB , where $|\overrightarrow{AD}| : |\overrightarrow{DB}| = 2:3$. Given

$\overrightarrow{AD} = \underline{a}$, $\overrightarrow{AC} = \underline{b}$ and $\overrightarrow{CB} = \underline{c}$, show that $\underline{b} = \frac{1}{2}(5\underline{a} - 2\underline{c})$.

$$\begin{aligned}\overrightarrow{DB} &= \frac{3}{2} \underline{a} \\ \overrightarrow{AC} &= \overrightarrow{AB} + \overrightarrow{BC} \\ \underline{b} &= \overrightarrow{AD} + \overrightarrow{DB} - \underline{c} \\ &= \underline{a} + \frac{3}{2} \underline{a} - \underline{c} \\ &= \frac{5}{2} \underline{a} - \underline{c} \\ &= \frac{1}{2} (5\underline{a} - 2\underline{c})\end{aligned}$$

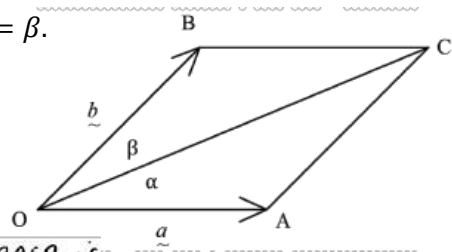


- 10 $OACB$ is a rhombus. $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$, $\angle AOC = \alpha$ and $\angle COB = \beta$.

- (i). For the vectors in the diagram above, use properties

of the dot product to prove $\underline{a} \cdot (\underline{a} + \underline{b}) = \underline{b} \cdot (\underline{a} + \underline{b})$

$$\begin{aligned}\text{LHS} &= \underline{a} \cdot (\underline{a} + \underline{b}) = \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} \\ &= |\underline{a}|^2 + \underline{a} \cdot \underline{b} = |\underline{b}|^2 + \underline{a} \cdot \underline{b} \quad \text{since } OACB \text{ is a rhombus} \\ &= \underline{b} \cdot \underline{b} + \underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{b} + \underline{b} \cdot \underline{a} \\ &= \underline{b} \cdot (\underline{b} + \underline{a}) = \underline{b} \cdot (\underline{a} + \underline{b}) = \text{RHS}\end{aligned}$$



- (ii). Hence, prove diagonal OC bisects $\angle AOB$.

$$\begin{aligned}\underline{a} \cdot (\underline{a} + \underline{b}) &= \underline{b} \cdot (\underline{a} + \underline{b}) \\ |\underline{a}| \times |\underline{a} + \underline{b}| \cos \alpha &= |\underline{b}| \times |\underline{a} + \underline{b}| \cos \beta \\ |\underline{a}| \times |\underline{a} + \underline{b}| \cos \alpha &= |\underline{b}| \times |\underline{a} + \underline{b}| \cos \beta \\ \cos \alpha &= \cos \beta \\ \alpha &= \beta \\ \therefore OC \text{ bisects } \angle AOB\end{aligned}$$

- 11 Given that $\underline{a} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$, find:

(i). $|\underline{a}|$ $|\underline{a}| = \sqrt{(-4)^2 + (1)^2} = \sqrt{17}$

(ii). $\underline{a} \cdot \underline{b}$ $\underline{a} \cdot \underline{b} = (-4) \times (-3) + (1) \times (5) = 17$

- (iii). The exact angle between $\underline{a}$ and $\underline{b}$

$$\begin{aligned}\underline{a} \cdot \underline{b} &= |\underline{a}| \cdot |\underline{b}| \cos \theta \\ \cos \theta &= \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| \cdot |\underline{b}|} \quad |\underline{b}| = \sqrt{(-3)^2 + (5)^2} \\ &= \frac{17}{\sqrt{17} \times \sqrt{34}} \\ \cos \theta &= \frac{1}{\sqrt{2}} \\ \theta &= \frac{\pi}{4}\end{aligned}$$

- (iv). $\text{proj}_{\underline{b}} \underline{a}$

$$\text{proj}_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{\underline{b} \cdot \underline{b}} \times \underline{b} = \frac{17}{34} \begin{bmatrix} -3 \\ 5 \end{bmatrix} = \begin{bmatrix} -3/2 \\ 5/2 \end{bmatrix}$$